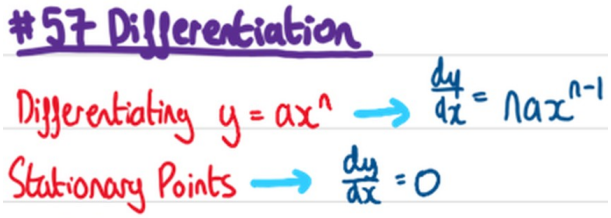
Worksheet 45 · Differentiation IGCSE

Edexcel IGCSE Mathematics · past paper questions

Name:	Subject: Math	Date:
Learning Objectives: Students will be able to: • Differentiate expressions of the form ax^n • Find the gradient of a curve at a point • Find turning points of a curve		
Grade: / 70	**Teacher's Feedback:**	

Answer in the space given. Show your working. Marks for each part are shown in brackets.

Key notes



Questions *Differentiation*

Question 1 (5 marks)

The curve C has equation $y = \frac{1}{3}x^3 - 9x + 1$

(a) Find $\frac{dy}{dx}$

$\frac{dy}{dx} =$ ..... (2)

(b) Find the range of values of x for which C has a negative gradient.

..... (3)

Question 2 (6 marks)

$y = x^3 - 2x^2 - 15x + 5$

(a) Find $\frac{dy}{dx}$

$\frac{dy}{dx} =$ ..... (2)

C is the curve with equation $y = x^3 - 2x^2 - 15x + 5$

(b) Work out the range of values of x for which **C** has a negative gradient.

..... (4)

Question 3 (7 marks)

A curve **C** has equation $y = x^3 - x^2 - 8x + 12$

(a) Find $\frac{dy}{dx}$

$\frac{dy}{dx} =$ ..... (2)

The curve **C** has two turning points.

(b) Work out the x coordinates of the two turning points.
Show your working clearly.

..... (3)

(c) Show that the x-axis is a tangent to the curve **C**.

..... (2)

Question 4 (5 marks)

The point A is the only stationary point on the curve with equation $y = kx^2 + \frac{16}{x}$ where k is a constant.

Given that the coordinates of A are $\left(\frac{2}{3}, a\right)$

find the value of a.
Show your working clearly.

$a =$ .....

Question 5 (8 marks)

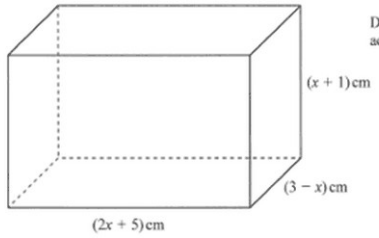


Diagram **NOT** accurately drawn

The diagram shows a cuboid of volume V cm³

(a) Show that $V = 15 + 16x - x^2 - 2x^3$

There is a value of x for which the volume of the cuboid is a maximum.

b) Find this value of x.
Show your working clearly.
Give your answer correct to 3 significant figures.

$x =$ .....

(5)

Question 6 (5 marks)

The diagram shows a solid cuboid.

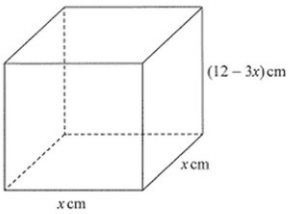


Diagram **NOT** accurately drawn

The total surface area of the cuboid is A cm²

Find the maximum value of A.

.....

Question 7 (6 marks)

$y = x^3 - 6x^2 - 15x$

(a) Find $\frac{dy}{dx}$

$\frac{dy}{dx} =$ .....

(2)

The curve with equation $y = x^3 - 6x^2 - 15x$ has two stationary points.

(b) Work out the coordinates of these two stationary points.

(..... , ...)

(..... , ...)

(4)

Question 8 (3 marks)

$ABCED$ is a five-sided shape.

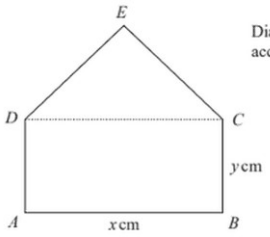


Diagram **NOT** accurately drawn

$ABCD$ is a rectangle.
CED is an equilateral triangle.

$AB = x$ cm $\quad BC = y$ cm

The perimeter of $ABCED$ is 100 cm.
The area of $ABCED$ is R cm²

(a) Show that $R = \frac{x}{4}\left(200 - \left[6 - \sqrt{3}\right]x\right)$

(3)

Question 9 (6 marks)

(b) (i) Find the value of x for which R has its maximum value.

Give your answer in the form $\frac{p}{q - \sqrt{3}}$ where p and q are integers.

$x =$ .....

(2)

(ii) Explain why the maximum value of R is given by this value of x.

.....

.....

.....

(1)

Question 10 (6 marks)

The curve **C** has equation $y = ax^3 + bx^2 - 12x + 6$ where a and b are constants.

The point A with coordinates $(2, -6)$ lies on **C**
The gradient of the curve at A is 16

Find the y coordinate of the point on the curve whose x coordinate is 3
Show clear algebraic working.

$y =$ .....

Question 11 (7 marks)

The diagram shows a sketch of part of the curve with equation $y = x^2 - \frac{p}{x}$ where p is a positive constant.

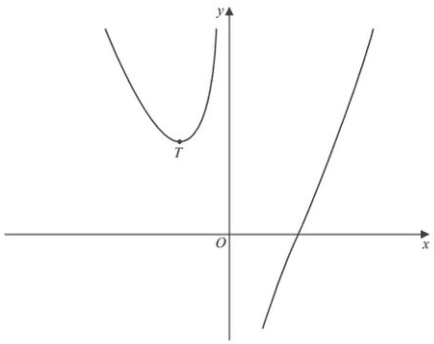


Diagram **NOT** accurately drawn

For all values of p, the curve has exactly one turning point and this turning point is a minimum shown as the point T in the sketch.

For the curve where the x coordinate of T is -3

(a) find the value of p

$p =$ .....

(4)

The line with equation $y = k$ is a tangent to the curve with equation $y = x^2 - \frac{16}{x}$

(b) Find the value of k

$k =$ .....

(3)

Question 12 (6 marks)

A solid, **S**, is made from a hemisphere and a cylinder.

The centre of the circular face of the hemisphere and the centre of the top face of the cylinder are at the same point.

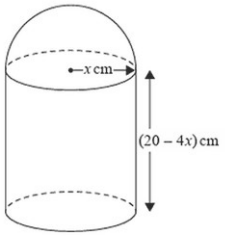


Diagram **NOT** accurately drawn

The radius of the cylinder and the radius of the hemisphere are both x cm.
The height of the cylinder is $(20 - 4x)$ cm.

The volume of **S** is V cm^3 where $V = \frac{1}{3}\pi y$

Find the maximum value of y.
Show clear algebraic working.